Final Design of Automated Paintbrush Cleaner
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Introduction and Objectives

The current process for artists looking to clean oil paint from paintbrushes can be **time consuming**, **skin-damaging**, and **dreadful**. It requires the artist to soak the brush in paint solvent, swipe it across their hand, and rinse with soap and water, which can take 1-2 minutes per brush.

So, our team set out to design a machine that can **clean many brushes** at a time with significantly **less maual labor** from the artist. In addition, this machine will **reduce contact** between the artist and the toxic solvent.

Final Design



Image 1: Final Prototype

The complete mechanism consists of a black enclosure housing all moving and electrical components which sits on top of a detachable acrylic cup. During the prototyping stage, we initially had a much bulkier structure, however we were able to slim it down substantially given the strength of the 3D print. Additionally, the resin used to hold the paintbrushes was changed to a less brittle and more flexible resin to enable faster brush insertion.

Design Components

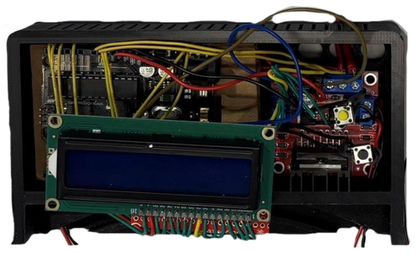


Image 2: Electrical Components



Image 3: Motor



Image 4: Gear and Brush Holder

The machine is powered with a 12V battery which users can recharge as needed. The circuitry includes two buttons – one for a cycle with solvent and one for a shorter cycle with soap and water – and a screen that shows which cycle is running (see image 2). Also powered through the circuitry are two motors with gears attached (see image 3) that alternate direction periodically throughout the cycle. Together, they create a rotational motion for the brushes which are housed in the large gear (see image 4). A silicone rubber holds brushes in their spot. When inserting brushes into the large gear, they should be arranged by size, and their heights should be adjusted to touch the bottom of the cup.



Image 5: Bottom Casing



Image 6: Base and Cup



Image 7: Telescoping Legs

The large gear sits inside of a casing which has ball bearings on both the top and the bottom to ensure that the gear can rotate smoothly (see image 5). The casing sits on top of a clear acrylic cup (see image 6) which holds the solvent or soap and water. Inside of the cup is a silicone mat with finger-like texture that can get into and clean the brushes without damaging them. After prototyping and receiving feedback from the client, telescoping legs were added to the sides of the casing (see image 7). The legs allow the user to raise the casing to easily slide the cup in and out.

Design Criteria and Testing

Design Criteria	Target Value	Testing Plan	Result
Ease of Use	Machine can be set up and run in under 5 minutes	Record the time it takes our peers to set up and run the machine after reading the instructions	Pass – it took approximately 2 minutes and 30 seconds to set up the machine and run
Aesthetics	Averages a score above a 3 on an aesthetics test	Use a survey with 3 categories - color, size, and shape - for peers to rate the machine on a scale of 1 to 5	Pass – average score is 3.84
Effectiveness	90% of paint is removed after one full cycle	Compare the color of the brush before being covered in paint and after being washed. Use a scale and panel of judges to determine the change of color	TBD – more testing must be conducted
Capacity	Machine can run at full capacity for 15+ cycles	Fill the machine with 18 brushes and run it	Pass – the machine can handle running with 18 brushes

Conclusion

We achieved our goal of helping our client James Keul automatize the paint brush cleaning process through our final design, which is a machine that includes an acrylic cylinder, textured based, rotating gear, and an interactive interface. We reached our final design by brainstorming, using Pugh screening and scoring matrixes to narrow down our options, creating low, medium, and high fidelity prototypes, and testing to find any issues and finalize our design. We remained in contact with our client, receiving feedback along the way. In the end, we have created a machine that successfully and efficiently cleans the brushes while accounting for all our design objectives and constraints.

Device Instructions

Acknowledgements

We thank the following for their support on our project:

- Dr. Rebecca Simmons, Ph.D, *Professor*
- Dr. Stefano Curtarolo, Ph.D, *Professor*
- Joey Flemming, *Teaching Assistant*
- Kevin Shores, *Technical Mentor*
- Ali Stocks, *Foundry Manager*
- James Keul, *Client*
- Duke Pratt School of Engineering